

02C0343 – TALENT4S3

Regional Brain Drain: Comparative Analysis of European Strategies for Talent Retention and Smart Specialization

November 1, 2024

Document written by University of Groningen (AP08):

Dr. Thijs Broekhuizen

Prof. dr. Sierdjan Koster

Dr. Agata Troost

Table of Contents

Introduction.....	3
1. Defining Brain Drain	4
2. Regional Brain Drain: Key Drivers of Migration and Retention.....	6
2.1 Economic factors	6
2.2 Educational opportunities	6
2.3 Quality of life and cost of living	6
3. Policy Approaches	7
3.1 Enhancing economic and social conditions in sending regions.....	7
3.2 Future-oriented approaches	7
3.3 Reframing immobility as an asset	7
3.4 Encouraging return migration	7
4. Brain drain in 7 European Regions	9
4.1 Partners of the TALENT4S3 project	9
4.2 Comparison criteria to evaluate strategies for talent retention and smart specialisation	9
4.3 Highlighting communalities and distinct factors.....	21
4.3.1 Communalities between regions.....	21
4.3.2 Differences between regions.....	21
5. Translation into Interregional Thematic Workshops	22
6. References	24

List of Tables

<i>Table 1: Comparison criteria to evaluate regional strategies</i>	<i>9</i>
<i>Table 2: Comparison of regional strategies of TALENT4S3 partners</i>	<i>11</i>
<i>Table 3: Proposed Interregional Thematic Workshops</i>	<i>22</i>

Introduction

The analysis of brain drain – the migration of relatively highly educated individuals from less developed to more developed regions or countries – in the European context is important because it addresses a critical challenge posed by the emigration of qualified individuals and the resulting regional disparities. The European Union, with its freedom of movement, creates a context in which regions have to deal with brain drain not only to more vital areas within the country but also in other parts of the continent. While the EU values labour mobility as a key to its competitiveness, the resulting brain drain in some regions may exacerbate regional imbalances as it accelerates ageing, depopulation and local work force shortages. The Smart Specialization framework intersects with the idea of brain drain in two contrasting ways. On the one hand, Smart Specialization Strategies offer a framework to counteract the negative effects of brain drain by focusing on regional strengths and fostering innovation. Through collaborative policy-making involving key stakeholders, regions can build ecosystems that support quality jobs and align workforce skills with market needs, thereby retaining and attracting talent essential for sustainable development.

On the other hand, effective S3 implementation relies on the availability of regional talent and competitive resources to support such a strategy. This is true for the definition and further development of ecosystems around certain specific economic activities and for the institutional context in which the Smart Specialization Strategy is formulated (Di Cataldo et al., 2021). In short, S3 offers the opportunity for regions to counteract brain drain and its effects. At the same time, its successful implementation also relies on the availability of local talent. Leveraging talent for the regional implementation of S3 is then crucial to breaking the self-reinforcing circle in the wake of brain drain processes.

This report, written for the TALENT4S3 project, has two objectives. The first objective is to provide a brief theoretical background to the concept of brain drain in a European context. The first section defines brain drain and highlights its benefits, like remittances and international knowledge sharing, as well as negative consequences, including uneven development and inequality. The second section describes important factors contributing to regional brain drain, including economic opportunities, access to education, quality of life, and cost of living. The third section summarises the relevant mechanisms and policies that institutions (e.g., regional development agencies, ministries of economic development, etc.) have at their disposal, including enhancing economic and social conditions to attract and retain talent, adopting future-oriented strategies that prioritize sustainable and knowledge-based sectors, reframing immobility as a strategic asset by strengthening local opportunities, and encouraging return migration through targeted incentives.

The second objective is to provide a comparison of the regional brain drain of seven partners of the TALENT4S3 project. Comparing the seven regions' similarities and differences is vital for this project, as it helps identify both common challenges and distinct opportunities in addressing brain drain. This understanding allows the project to create (tailored) strategic recommendations, facilitate mutual learning between regions, and design more effective, context-specific policies for talent retention and attraction within the framework of Smart Specialisation Strategies.

1. Defining Brain Drain

Brain drain refers to the international transfer of resources in the form of human capital. It applies to the migration of relatively highly educated individuals, particularly professionals such as scientists, engineers, doctors, and academics, from one region or country to another (Beine et al., 2008). While this phenomenon is often associated with international migration from developing to developed countries (Docquier & Rapoport, 2012), brain drain also occurs within national boundaries, manifesting as regional brain drain. This internal migration typically involves professionals moving from less developed or rural areas to richer or more urbanised regions within the same country. Migration between different regions of the European Union can be seen as a special case because of the freedom of movement of EU citizens, which may result in extra pressure on the regions dealing with the emigration of skilled individuals.

Docquier & Rapoport (2012) outline several waves of academic interpretation of the international brain drain. Authors from the first wave in the 1960s emphasised the assets and remittances from the emigrants, which can compensate the sending countries for any actual loss caused by brain drain (Berry & Soligo, 1969), as well as high-skill migrants' contribution to knowledge, an international public good (Grubel & Scott, 1966). The second wave, starting in the 1970s, was more critical of the brain drain's contribution to increased inequality at the international level, with rich countries becoming richer at the expense of poor countries (Bhagwati & Hamada, 1974). The third wave, emerging in the 1990s due to the availability of new data sources allowing for extensive empirical testing, provides – as Docquier & Rapoport (2012) argue – a more balanced view of brain drain, providing some benefits to the sending countries.

In their case study of brain drain in Europe, Docquier & Rapoport (2012) start with comparison statistics showing that, despite producing more PhD level graduates than USA, European countries have fewer researchers per capita and experience outmigration of top skilled workers to the United States, Australia, Canada, and New Zealand, which is not fully compensated by the immigration from developing countries (p. 714). The intra-EU migration, on the other hand, is characterised mainly by the “brain gain” of richer Western European countries from Southern and Eastern Europe. However, there are also authors arguing for the presence of positive externalities, such as higher educational attainment, in the sending countries (Mayr & Peri, 2009). Teney (2021) situates the migration of physicians from different European regions to Germany on a scale between brain gain (for Germany) and brain circulation. Brain circulation, in contrast to the asymmetrical brain gain/drain, is a scenario in which “the intra-European migration of highly skilled workers would represent a labour response to the mismatch between skills and regions within the EU: the immigration of highly skilled Europeans would be a reaction to a differential in labour market demand, and its duration would depend on how the demand for skills developed across different regions” (p. 70). The migration of physicians to Germany from North-Western Europe comes close to the brain circulation scenario; South-Western European physicians would be a bit closer to brain drain, followed by physicians from the 2004 EU member states and ending with physicians from the post-enlargement member states, nearest to the brain gain scenario. This hierarchy corresponds to the economic differentials between these regions of origin and Germany. Physicians are more likely to migrate permanently in the brain gain/drain scenario and be motivated by economic push-pull factors than those in the brain circulation scenario, motivated by personal reasons such as relationships and proximity to their country of origin.

Next to the international intra-EU migration, less developed regions within the EU states face extra competition in the European common market. Although research on inter-regional brain drain seems relatively scarce, some studies and analyses (Artz, 2003; Faggian et al., 2018; Vazzana & Rudi-Poloshka, 2019; Zhou et al., 2018) in the European spatial context (Biedka et al., 2022; Haapanen & Tervo, 2012; Mærsk et al., 2021; Venhorst et al., 2011) exist. There is also a body of literature on “left-

behind places,” with detailed insights into the labour markets of European regions (Dobrzanski et al., 2024). This recent current of literature explores themes relevant to brain drain. However, it is inspired mainly by interest in less economically developed regions showing political support for populist and sometimes extremist initiatives (MacKinnon et al., 2022). In this sense, the policy and empirical analyses of left-behind places extend beyond the focus on high-skilled workers, typical for policies and studies concerning brain drain, and aim to find solutions to improve the situation of all inhabitants of a region (Fiorentino et al., 2024). Still, these are usually also highly relevant to the opportunities and quality of life of skilled (possible) migrants. Therefore, the following overview of regional brain drain processes includes insights from specific brain drain literature and works on left-behind places.

2. Regional Brain Drain: Key Drivers of Migration and Retention

2.1 Economic factors

In the European context, migration of human capital usually favours large metropolitan areas (Biedka et al., 2022). That is consistent with the fundamental economic factors behind migration decisions: employment opportunities – the availability of jobs matching one's qualifications – and wage premium. Therefore, an opportunity-rich labour market is of key importance in attracting or retaining highly skilled workers, especially recent graduates (Venhorst et al., 2011). Rural areas do not necessarily have to be economically underdeveloped; Dobrzanski et al. (2024) found no relationship between the share of agriculture in total employment and the productivity level of the region in Central and Eastern European regions. Still, mismatches can emerge more easily between the skills of the local population and those needed by the local economy.

2.2 Educational opportunities

Developed and urban regions often offer better educational programmes and institutions. Many of their students remain in the region after graduation. Also, working professionals in less developed areas may face limited access to further education and professional development, leading them to migrate in search of better training and career advancement opportunities. Even regions with their own higher education can experience student outmigration, as mobility can be seen as helping young people progress in life (Mærsk et al., 2021).

2.3 Quality of life and cost of living

Next to economic and educational factors, high quality of life – lack of pollution, good spatial planning, access to culture, nature, healthcare, childcare, and sports facilities – can motivate people to stay in or move to a region. Another significant factor is the cost of living, especially housing (Venhorst et al., 2011). Affordable housing becomes more critical with the current housing crisis in metropolitan areas; in general, rising prices of food and amenities can be perceived as insufficiently compensated by higher wages in high-cost-of-living areas. At the same time, underdeveloped regions may lack suitable housing or care facilities, even if general costs are lower.

3. Policy Approaches

3.1 Enhancing economic and social conditions in sending regions

Local governments should focus on creating a conducive environment for professional growth by offering competitive salaries, improving working conditions, and ensuring access to modern technology (Fiorentino et al., 2024; Venhorst et al., 2011). Additionally, investing in housing and enhancing the quality of life will make staying in these regions more appealing for skilled workers. Especially investing in affordable, sustainable, and attractive housing stock can contribute to regions' attractiveness in the context of European cities' widespread housing affordability crisis, with urban centers often inaccessible to outsiders despite their variety of job prospects (Arundel & Lennartz, 2020).

Fiorentino et al. (2024) argue that while the EU Cohesion Policy (CP) has been effective in creating employment opportunities in sectors such as innovation and transport infrastructure and effective in reducing inequalities between the EU member countries, it has been less successful in addressing the underdevelopment of within-country regions. A localized, diversified, evidence-based approach that aligns with regional strengths and needs, especially in sectors with growth potential, can help to reduce the migration of skilled workers to economic centers and improve the local economy.

3.2 Future-oriented approaches

Addressing brain drain requires a long-term, integrated strategy that spans multiple levels of governance. Lessons from German reunification, which included long-term fiscal transfers and resource allocation to lagging regions, demonstrate the importance of sustained commitment and resources in reducing regional inequalities (Fiorentino et al., 2024). Yet German reunification can also be an example of over-reliance on traditional industries. According to the EU Cohesion Policy priorities of the digital and green transformation, a future-oriented approach should emphasize knowledge-based sectors and higher-value industries, particularly in cities and regions with growth potential. This requires prioritising sustainable production and services. Local governments should ensure that investments modernize infrastructure and build capacity in high-value sectors that can sustain long-term growth and retain skilled workers.

Remote work offers new possibilities for employment in diverse sectors outside of traditionally economically dense regions. Still, expanding remote work practices should not conflict with employees' work-life balance.

3.3 Reframing immobility as an asset

Rather than viewing outmigration as inevitable, peripheral regions should explore ways to retain young professionals and graduates by reframing immobility as a strategic and advantageous choice (Mærsk et al., 2021). Investing in high quality local higher education can be particularly helpful. Encouraging regional educational institutions and businesses to collaborate can create opportunities for local talent, reducing the pressure for young people to relocate to urban centers for career progression. Viewing immobility as a viable development strategy can unlock new potential for regional growth.

3.4 Encouraging return migration

Many countries have introduced policies encouraging skilled professionals who have migrated abroad to return home (Beine et al., 2008). These initiatives provide incentives for returnees, such as tax breaks, research grants, and investment opportunities. Similar policies could be considered by local governments experiencing a regional brain drain, which emphasises connecting the new expertise of inhabitants who studied or worked in other regions with regional strengths and traditions. These policies are explicitly geared towards trying to influence the decision to migrate back to the home region. A complementary approach may be to try and sustain the network of regional 'alumni' and keep them

connected to the home region. Any policies geared towards influencing the actual migration decision are likely more effective. In addition, while talent is living in another region, some of their skills, experiences, and networks may still be leveraged for local development if regions manage to keep a relationship with those who have left. Conceptually, such an understanding of brain drain can be seen as an alternative take on the original idea of remittances. In this understanding of the idea, brain drain is not a means to initiate a counter-stream of finances but rather of knowledge and pertinent network relations.

4. Brain drain in 7 European Regions

4.1 Partners of the TALENT4S3 project

The TALENT4S3 project focuses on improving policies related to retaining and attracting highly skilled talent to ensure the effective implementation of Smart Specialisation Strategies (S3) in European regions. The project addresses the challenge of brain drain and regional disparities by creating tailored strategies for talent retention and attraction, thus boosting regional innovation and economic development.

The seven actors active in the TALENT4S3 project represent national and regional authorities and public administration bodies that can influence the region's brain drain through policy making. The University of Groningen (AP08) has asked these actors to fill in a document representing their State of the Art. These documents have been compared to address the common and unique regional challenges related to talent migration. A comparison of the seven regions' commonalities and differences is crucial for the TALENT4S3 project as it enables the identification of shared challenges and unique opportunities in tackling brain drain. By understanding these variations, the project can tailor strategic recommendations, foster mutual learning among regions, and develop more effective and context-specific policies for talent retention and attraction within the framework of Smart Specialisation Strategies.

4.2 Comparison criteria to evaluate strategies for talent retention and smart specialisation

The following criteria are identified to evaluate the seven European regions.

Table 1: Comparison criteria to evaluate regional strategies

Criterion	Description
1. Contextualization and Policy Instrument Description	Provides an overview of the region's economic and demographic context, emphasizing its role in national and European frameworks. Describes specific policy instruments targeting brain drain and their alignment with Smart Specialisation Strategies, along with the available data infrastructure.
2. Regional Background Information	Outlines the demographic and human capital profile of the region, focusing on population, migration patterns, education levels, and the overall brain drain situation. Discusses the policy priority of addressing brain drain.
3. Analysis of Brain Drain Factors	Identifies key factors driving brain drain, such as job availability, housing conditions, and social environment. Analyses how these factors impact the migration of skilled professionals.
4. Evaluation of Brain Drain Consequences	Assesses the effects of brain drain on socio-economic conditions, human capital, and competitiveness. Considers the impact on key sectors like education, health, and industry, as well as economic and social indicators.
5. Study of Existing Policies and Activities	Reviews current policies and initiatives aimed at retaining and attracting talent. Identifies gaps, challenges, and the effectiveness of these measures, while also evaluating policy maturity, data use, and stakeholder involvement.

6. Conclusions	Summarizes key findings, highlights challenges, and suggests strategies to address brain drain. Emphasizes the need for policy improvements to promote sustainable regional growth.
7. Uniqueness of region	Summarizes the brain drain situation of the region, and highlights the unique aspects of the region.

Table 2: Comparison of regional strategies of TALENT4S3 partners

Section	Lithuania (LIC)	Emilia-Romagna (RER)	Podravje	Extremadura (FUNDECYT)
1. Contextualization and Policy Instrument Description	Lithuania has implemented policies focused on both attracting foreign talent and retaining local talent, with a specific emphasis on educational and economic incentives. The country has recognized the challenges posed by significant emigration, particularly among the younger population, and has aimed to create a more conducive environment for retaining this talent through the improvement of living conditions, professional opportunities, and targeted incentives for return migration. The policy instruments include enhanced funding for higher education, scholarships to keep students in the country, and collaborations with international institutions to boost local education standards. Additionally, efforts are made to integrate these policies with the broader Smart Specialisation Strategy (S3) to align talent retention with national economic goals.	The Emilia-Romagna region addresses talent attraction and retention through Regional Law 14 February 2023 - n. 2, which is specifically designed to attract and retain highly specialized talents. This law is embedded within a broader strategy that integrates with the Smart Specialisation Strategy (S3) 2021-2027, emphasizing innovation, research, and international collaboration. The policy includes support for local businesses, universities, and research centers to create a comprehensive ecosystem that fosters talent development. It also offers incentives for talents abroad to relocate to Emilia-Romagna, providing them with opportunities for professional growth, access to high-quality public services, and a supportive living environment. The region's focus on high-tech sectors, such as automotive, agriculture, and tourism, plays a crucial role in attracting global talent.	Podravje's approach to preventing brain drain primarily revolves around enhancing economic opportunities within the region. The policies are closely tied to regional economic development strategies that aim to create jobs, improve local infrastructure, and enhance the overall quality of life. A key focus is on boosting the region's attractiveness to both retain local talent and attract professionals from other regions or countries. Initiatives include improving educational facilities, offering professional training programs aligned with market needs, and providing financial incentives for businesses that employ local youth. Additionally, the region has initiated several public-private partnerships aimed at fostering innovation and entrepreneurship as part of the broader S3 objectives.	Extremadura has focused its policies on improving education and employment opportunities to mitigate the high levels of youth emigration. The region's strategy is aligned with the Smart Specialisation Strategy (S3), emphasizing sustainable growth and the creation of new economic sectors that can offer attractive job opportunities for young people. Policies include investment in higher education, vocational training, and entrepreneurship programs designed to provide local talent with the skills needed in the regional labor market. Moreover, the region is working on improving its infrastructure and connectivity to make it a more attractive place to live and work. These efforts are complemented by targeted campaigns to promote Extremadura as a region of opportunities for young professionals and innovators.

2. Regional Background Information	Lithuania faces significant demographic challenges, characterized by a declining and aging population. High rates of out-migration, particularly among young people seeking better educational and employment opportunities abroad, have led to a substantial loss of human capital. This demographic shift has created challenges for regional development, with a shrinking workforce and an increasing dependency ratio. The country is focusing on improving its higher education system and aligning it more closely with labor market needs to address these issues.	Emilia-Romagna is one of Italy's most economically prosperous regions, with a diverse economy that includes strong sectors in manufacturing, automotive, agriculture, and tourism. The region's population of about 4.4 million is relatively stable, but it is aging, with a lower proportion of young people. This demographic trend poses challenges for future economic growth, as the region needs to attract and retain young talent to maintain its competitive edge. Despite these challenges, Emilia-Romagna remains a key player in both national and European contexts, contributing significantly to Italy's GDP and exports.	Podravje is experiencing economic challenges that have led to significant out-migration, particularly among the youth. The region's demographic profile is marked by a declining and aging population, with many young people leaving in search of better opportunities. This trend negatively impacts the region's economic prospects, as the loss of young, skilled workers hinders local development efforts. Efforts are being made to improve local educational and economic opportunities, but the challenge remains substantial.	Extremadura is one of Spain's less developed regions, facing high youth unemployment and limited economic opportunities, which have contributed to significant emigration. The region's demographic profile is characterized by an aging population and a high dependency ratio, with many young people leaving for better prospects elsewhere. This trend poses serious challenges to the region's sustainable development. Extremadura is attempting to leverage its rich cultural heritage and natural resources to create new economic opportunities and retain its young population.
------------------------------------	---	--	--	--

3. Analysis of Brain Drain Factors	Brain drain in Lithuania is primarily driven by economic instability and limited career opportunities, particularly in high-skill sectors like IT, engineering, and healthcare. Young, educated Lithuanians often leave the country for better job prospects and higher salaries in Western Europe. The quality of life, including access to services and amenities, also influences this out-migration. Despite efforts to improve the local economy and job market, Lithuania continues to struggle with retaining its talent, particularly in sectors crucial to its economic future.	While Emilia-Romagna is generally attractive due to its high quality of life and robust job market, the region faces challenges in retaining highly skilled professionals. The limited availability of specific high-skill job opportunities, especially in emerging sectors, leads some professionals to seek better career prospects elsewhere. Additionally, the aging population and demographic shifts create a competitive environment for retaining and attracting young talent. Despite these issues, Emilia-Romagna remains a net attractor of talent within Italy, particularly from less economically developed southern regions.	Podravje faces significant brain drain challenges, driven by a lack of economic opportunities and limited educational and professional prospects. The region's developing economic structure does not offer enough high-quality jobs to retain its educated youth. Many young people leave for better opportunities in larger urban centers or abroad, particularly in sectors like IT, engineering, and finance. The perceived lack of cultural and social opportunities also contributes to this out-migration, making it difficult for the region to retain its young talent.	Extremadura's brain drain is largely a result of economic underdevelopment and the limited availability of high-skill jobs. High levels of youth unemployment drive many young people to seek opportunities elsewhere, particularly in more developed regions of Spain or abroad. The lack of dynamic industries and innovation capacity further exacerbates the problem, as the region struggles to create an environment conducive to retaining or attracting talent. Social factors, including limited cultural and leisure activities, also contribute to the perception that Extremadura offers limited prospects for a fulfilling life, prompting more people to leave.
---	---	---	---	--

4. Evaluation of Brain Drain Consequences	The consequences of brain drain in Lithuania are severe, leading to a significant shortage of skilled labor, which hampers economic growth and innovation. This shortage creates a vicious cycle, where the lack of opportunities drives more emigration, further weakening the local economy. The social consequences are also profound, as communities become increasingly aged and depopulated. This demographic shift affects local services, education, and overall quality of life. Lithuania faces an urgent need to create more opportunities for young people and improve living conditions to retain its talent and ensure sustainable development.	In Emilia-Romagna, the brain drain primarily impacts the region's economic development by reducing the availability of highly skilled professionals in key sectors such as manufacturing and technology. The loss of specialized talent weakens the region's competitiveness and innovation capacity, which are critical for maintaining its economic vitality. Additionally, the aging population, coupled with the outflow of younger, educated individuals, poses a significant challenge to the region's long-term sustainability. This growing gap between the demand for high-skill jobs and the available talent pool threatens to slow down economic growth and innovation in the region.	The outflow of young, skilled workers in Podravje has significant negative consequences for the region's economic prospects. The loss of talent hinders local development efforts, as there is a growing mismatch between the skills needed by the region's economy and the available workforce. This brain drain exacerbates the region's economic challenges, making it more difficult to attract investment and drive growth. The persistent loss of young professionals also impacts social structures, leading to aging communities and reduced dynamism in the region.	Extremadura suffers from severe consequences due to brain drain, particularly in terms of economic and social sustainability. The continued out-migration of young, educated individuals weakens the region's workforce, limiting its ability to develop and sustain key industries. The loss of talent further exacerbates the region's economic underdevelopment, making it difficult to attract new businesses or foster innovation. Socially, the brain drain contributes to an aging population and depopulation in rural areas, reducing the vibrancy and cohesion of communities. These challenges highlight the critical need for targeted interventions to retain and attract talent to the region.
--	--	--	---	---

5. Study of Existing Policies and Activities	Lithuania has implemented a range of policies aimed at retaining and attracting talent, with a strong focus on improving educational infrastructure and creating economic opportunities. These policies include enhancing higher education, providing scholarships to retain students within the country, and offering incentives for return migration. The government has also sought to align these efforts with the Smart Specialisation Strategy (S3), which aims to foster innovation and economic growth. However, the effectiveness of these policies is still under evaluation, as brain drain continues to pose a significant challenge.	Emilia-Romagna has developed a comprehensive set of policies to address brain drain, particularly through Regional Law 14 February 2023 - n. 2. This law is focused on attracting, retaining, and valorizing highly specialized talents by supporting local businesses, universities, and research centers. Key activities include providing research grants, promoting international collaborations, and enhancing local services to make the region more attractive to talent. The region also emphasizes data-driven decision-making, with strong monitoring and evaluation mechanisms in place to assess the impact of these policies. The integration with the Smart Specialisation Strategy (S3) ensures that these efforts are aligned with broader economic goals.	In Podravje, existing policies are primarily economic, aiming to create job opportunities and improve local infrastructure to retain talent. The region has focused on enhancing education and professional training programs that are aligned with market needs. Public-private partnerships are being promoted to foster innovation and entrepreneurship. However, the effectiveness of these policies is limited by the region's economic challenges, and there is a need for more targeted interventions to address the root causes of brain drain.	Extremadura's policies are centered around improving education, vocational training, and employment opportunities to retain its young population. The region has invested in enhancing the quality of its higher education institutions and has implemented programs to support entrepreneurship and innovation. Additionally, Extremadura has worked on improving its infrastructure and connectivity to make the region more attractive to both residents and potential migrants. However, the region faces significant challenges in fully reversing the brain drain trend, and there is a continued need for more effective policy interventions that can address the underlying economic and social factors driving emigration.
---	--	---	--	---

6. Conclusions	The conclusions for Lithuania emphasize the urgent need to address the ongoing brain drain that continues to deplete the country's talent pool. The report underscores that while efforts have been made to improve educational infrastructure and economic opportunities, these measures have not yet sufficiently stemmed the outflow of skilled professionals. The persistence of brain drain highlights the need for more robust and targeted policies that not only retain local talent but also attract those who have emigrated back to Lithuania. The report calls for enhanced coordination between educational institutions, government bodies, and the private sector to create a more attractive environment for young professionals.	Emilia-Romagna's conclusions reflect a twin challenge: the need to retain high-skilled talent within the region and the importance of attracting international professionals to maintain the region's competitive edge. The report emphasizes that while the region's policies, particularly the Regional Law 14 February 2023 - n. 2, are well-structured and aligned with the Smart Specialisation Strategy (S3), there is still a need for continuous innovation and adaptation. The region must remain vigilant in monitoring the effectiveness of its policies, ensuring that they are responsive to changing economic conditions and the evolving needs of the labor market.	The conclusions for Podravje highlight the region's ongoing struggle with brain drain, which poses a significant threat to its economic development. Despite efforts to enhance education and create job opportunities, the region continues to lose its young, skilled workforce to larger cities and other countries. The report suggests that more targeted interventions are needed to address the root causes of brain drain, such as improving the region's economic attractiveness and providing better career prospects for young professionals. The region must also strengthen its efforts to retain local talent by fostering a more supportive and dynamic environment.	Extremadura's conclusions stress the critical importance of reversing the brain drain trend to ensure the region's sustainable development. The report highlights the need for a multifaceted approach that not only improves economic opportunities and education but also enhances the overall quality of life in the region. While current policies have laid a foundation for addressing brain drain, the report calls for more innovative and integrated solutions that can better respond to the complex challenges facing the region. There is a particular emphasis on the need for collaboration across sectors and levels of government to create a more cohesive and effective strategy for retaining and attracting talent.
----------------	--	---	--	--

7. Uniqueness of region	Brain Drain Situation: • Significant out-migration of young, educated individuals, particularly to Western Europe. • Driven by economic instability, limited career opportunities, and better prospects abroad. • Efforts are being made to enhance education and align it with labor market needs, but challenges remain in retaining talent. Unique Aspect: • Lithuania faces a severe demographic challenge due to both a declining birth rate and high emigration, leading to a rapidly aging population.	Brain Drain Situation: • Despite being an economically prosperous region, Emilia-Romagna faces challenges in retaining highly specialized talent. • The region attracts talent from other parts of Italy but struggles to keep young professionals in emerging high-skill sectors. • A comprehensive legal and policy framework (Regional Law 14 February 2023 - n. 2) is in place to address these issues. Unique Aspect: • The region's strong economic base and integration with European markets make it a net attractor of talent within Italy, particularly from the less developed southern regions.	Brain Drain Situation: • Significant out-migration due to economic challenges and limited local opportunities. • Many young people leave for better prospects in larger urban centers or abroad, particularly in IT, engineering, and finance sectors. • The region is working to improve educational and economic opportunities, but the impact of these efforts is still limited. Unique Aspect: • Podravje's struggle is deeply tied to its economic underdevelopment, making the region particularly vulnerable to the loss of young, skilled workers.	Brain Drain Situation: • High levels of youth unemployment and limited economic opportunities have led to significant emigration. • The region suffers from an under-developed economy and lacks dynamic industries, which exacerbates the brain drain. • Efforts are being made to improve infrastructure and education, but more targeted interventions are needed. Unique Aspect: • Extremadura's brain drain is particularly severe due to its status as one of Spain's less developed regions, with a high dependency on its natural resources and cultural heritage for economic growth.
--------------------------------	---	---	--	--

Continuation of Table 2

Section	Lapland	Southern Ireland	North-West Romania
1. Contextualization and Policy Instrument Description	Lapland is a sparsely populated region in northern Finland, essential for its natural resources and strategic importance in Finland's green transition and the European Arctic. Its Sustainable Smart Specialisation Strategy (S3) 2023-2027 focuses on expertise renewal, international cooperation, and investments in green technology and sustainable energy.	The Southern Region of Ireland covers 42% of the country and includes key urban centers like Cork, Limerick, and Waterford. The region's strategic plan, the Regional Spatial and Economic Strategy (RSES), emphasizes Smart Specialisation, clustering, and creating a Learning Region to enhance human capital and economic prosperity.	The North-West region, encompassing six counties, focuses on innovation, industrial transition, and entrepreneurship through the North-West Regional Program 2021-2027. This policy instrument aligns with Smart Specialisation Strategies (S3) by supporting SMEs and promoting skills development.
2. Regional Background Information	The region has around 175,679 people, with an aging population and significant out-migration of young people. Despite challenges, tourism and green technology are growing industries, and internal migration has stabilized population decline. Urban centers like Rovaniemi are growing, but many rural areas continue to face depopulation.	The Southern Region has a population of over 1.7 million, with key sectors including technology, pharmaceuticals, and agriculture. The population is aging, with notable regional differences in population growth and density. Cork and Waterford are significant employment hubs, though housing shortages and infrastructure challenges persist.	North-West Romania is an economically diverse region with a population of 2.5 million. Key sectors include manufacturing, IT, and education, centered around urban hubs like Cluj-Napoca. The region is multicultural, with urban growth contrasting rural depopulation.
3. Analysis of Brain Drain Factors	Lapland faces brain drain due to its remote location, limited academic opportunities in technology and sciences, and better job prospects in southern Finland. The most affected sectors include healthcare, social services, and technology. The region also struggles to retain international talent despite	The Southern Region experiences brain drain due to housing shortages, limited regional higher education options, and the pull of larger urban centers like Dublin. The region's rural and coastal areas are particularly vulnerable, and sectors such as construction, agriculture, and IT struggle to attract and retain talent.	Brain drain in North-West Romania is driven by salary disparities with Western Europe, limited career opportunities, and inadequate infrastructure in key sectors like IT, medicine, and R&D. Migration is predominantly international, with popular destinations being Italy, Spain, and France.

	attracting seasonal workers in tourism and green energy.		
4. Evaluation of Brain Drain Consequences	Brain drain in Lapland exacerbates labor shortages, especially in public services and industries like healthcare, education, and mining. The loss of young talent increases the gap between Lapland and more prosperous regions in Finland, making it harder to sustain economic growth and public services.	The Southern Region's brain drain weakens its workforce and economic competitiveness, especially in key sectors like IT, construction, and education. Housing availability and affordability are significant barriers to retaining and attracting talent, and this issue is impacting regional economic development.	Brain drain exacerbates workforce shortages in IT, medicine, and education, reducing regional competitiveness and innovation. This contributes to an aging population in rural areas, worsening economic disparities between urban and rural communities.
5. Study of Existing Policies and Activities	Lapland has developed a joint regional alliance to address workforce shortages, focusing on retaining local youth, recruiting international talent, and building collaborations between education institutions and businesses. However, policies face challenges in meeting demand for both skilled and unskilled labor.	The Southern Region has implemented the Regional Skills Fora and cluster initiatives, such as the iEdHub and SME taskforces, to address skill gaps and enhance talent retention. However, housing shortages and under-investment in higher education remain significant policy challenges.	The North-West region's policies focus on industrial transition, SME support, innovation hubs, and entrepreneurship, with a specific emphasis on retaining local talent and attracting external experts. The region struggles with outdated infrastructure and low investment in R&D.
6. Conclusions	The region must attract both skilled and unskilled workers to maintain public services and drive innovation in key sectors. Lapland's low R&D investment limits its ability to compete with other regions, and better integration between education and business is needed to retain talent.	The Southern Region faces significant brain drain challenges due to housing and cost-of-living issues. While the region has strong economic sectors, addressing infrastructure gaps, especially in housing and higher education, is crucial for retaining talent and maintaining regional competitiveness.	North-West Romania must address salary disparities, improve infrastructure, and invest in R&D to retain and attract skilled professionals. Key challenges include rural depopulation and the migration of highly educated individuals.

7. Uniqueness of region	Brain Drain Situation: • Lapland, located in northern Finland, faces significant out-migration of young people due to its remote location and limited academic and career opportunities, particularly in the technology and sciences sectors. • The region struggles to retain talent in sectors such as healthcare, social services, and technology. • International talent, especially those drawn to seasonal jobs in tourism and green energy, is also difficult to retain long-term. • Population aging and depopulation in rural areas are critical challenges, even as urban centers like Rovaniemi see modest growth. Unique Aspect: • Lapland's harsh climate, geographical remoteness, and strategic importance for Finland's green transition set it apart. These factors uniquely position the region to focus on sustainability, but also make it particularly vulnerable to brain drain.	Brain Drain Situation: • Southern Ireland, despite being a relatively prosperous region with urban centers like Cork and Limerick, suffers from significant brain drain driven by housing shortages and a high cost of living, particularly in areas with economic growth. • Limited regional higher education options contribute to the migration of young people to Dublin and abroad, with sectors like IT, construction, and agriculture struggling to retain talent. • The region's rural and coastal areas are particularly vulnerable to depopulation. Unique Aspect: • Southern Ireland's housing crisis is a distinct factor, creating significant barriers to talent retention. This challenge is compounded by high costs of living, which is rather unique compared to the less-developed regions (e.g. Podravje and Extremadura).	Brain Drain Situation: • The North-West region of Romania, especially urban centers like Cluj-Napoca, faces significant out-migration, particularly of highly educated individuals in fields such as IT, healthcare, and education. • This brain drain is primarily driven by salary disparities between Romania and Western Europe, limited career growth opportunities, and outdated infrastructure in key sectors like research and development (R&D). • Migration is predominantly international, with popular destinations being Italy, Spain, and France. • Rural depopulation is another key challenge, with many young people leaving less developed rural areas in favor of urban hubs or foreign countries. Unique Aspect: • Strong IT Sector: North-West Romania, particularly Cluj-Napoca, is unique for its strong and rapidly growing IT sector. Despite this strength, the region struggles to retain highly skilled professionals who leave for better opportunities abroad due to salary gaps and insufficient infrastructure.
--------------------------------	---	--	--

4.3 Highlighting communalities and distinct factors

After analysing Table 2, the following communalities and differences are identified.

4.3.1 Communalities between regions

- **Economic Challenges:** All regions face economic challenges that drive brain drain, with young talents leaving in search of better job opportunities and higher salaries. Economic disparities between urban and rural areas (or between countries, see International Migration) are a strong influencer of migration. Many regions also face additional challenges, including aging populations and rural depopulation.
- **Insufficient Infrastructure & Limited R&D Investments:** All regions face economic challenges related to “infrastructure,” whether it is housing (Southern Ireland), limited access to higher education (Lithuania, Southern Ireland), limited or outdated physical infrastructure (Lapland, North-West Romania), or lack of economic diversification (Podravje, Extremadura). All regions simultaneously struggle with low investment in R&D and low innovation rates.
- **Importance of Education:** Each region recognizes the importance of aligning education with labor market needs to retain talent. Most regions aim for highly educated talents; Lapland aims to retain both high and lower-educated talents.
- **Need for Policy Interventions:** All regions have implemented or are developing policies aimed at retaining and attracting talent, though the effectiveness of these policies varies.
- **International Migration:** Some regions (Lithuania, Southern Ireland, and North-West Romania) have a significant portion of talent migrating internationally to countries in Western Europe in search of better career prospects and higher salaries abroad.

4.3.2 Differences between regions

- **Lithuania:** Demographic decline and aging population exacerbated by brain drain.
- **Emilia-Romagna:** Strong economic base but challenges in retaining high-skill talent in emerging sectors.
- **Podravje:** Economic underdevelopment makes it particularly vulnerable to brain drain.
- **Extremadura:** Severe brain drain due to an underdeveloped economy and high youth unemployment, with a reliance on natural resources for growth.
- **Lapland:** The region's geographical remoteness and strategic role in Finland's green transition make it unique. Despite its challenging climate and isolation, it needs to attract workers for public services and key sectors like green technology.
- **Southern Ireland:** The housing crisis and high cost of living create unique challenges in retaining talent, particularly in urban hubs like Cork and Limerick.
- **North-West Romania:** The region of Cluj-Napoca has the potential to become a strong economic and IT hub because it hosts numerous multinational IT companies. Despite this potential, the region continues to face brain drain challenges due to wage gaps with Western Europe, limited infrastructure, and an insufficient local innovation ecosystem.

5. Translation into Interregional Thematic Workshops

The selection of relevant themes for upcoming Interregional Thematic Workshops is based on the degree to which these themes are (a) common and relevant to most partners of TALENT4S3, and (b) can be influenced by the actors. The four following preliminary thematic workshops have been identified.

Table 3: Proposed Interregional Thematic Workshops

Time and Location	Interregional Thematic Workshop	Description	Goal
November 2024 , Spain, Extremadura	1. Brain drain Intelligence	This workshop discusses the data that is ideally available to assess the brain drain situation properly. This Brain Drain Intelligence is an important condition for an evidenced-based approach to address brain drain dynamics effectively. The University of Groningen (AP08) will showcase the 'Talent monitor' which was set up for this purpose in the North of the Netherlands.	To inspire the TALENT4S3 partners to reflect on the current data infrastructure available and the next steps to improve Brain drain Intelligence.
May 2025 , North-West Romania	2. Countering Brain drain - Housing and Amenities	The regional housing situation is an important condition for retaining talent that intends to stay (Brain Retain). In comparison to other retention factors—the availability of jobs, for example—governments have a relatively large scope to influence the housing market, and the provision of amenities. As such, it can be an important policy instrument.	To define the scope of investing in appropriate and accessible housing & amenities as a means to address Brain drain.
November 2025 , Finland, Rovaniemi	3. Long-term Regional Attachment and Return Migrants	Brain drain is not only problematic. It is also an opportunity for young talent to develop essential skills for S3. They can be ambassadors of the region, and some of them may come back with experiences and newly gained skills to benefit the regional economy. With that, it can be an opportunity for the region, too. To maximize this potential of brain drain, it is vital to structurally keep talent that remains involved with the region, so that the way back remains a viable opportunity.	To inspire a regional alumni policy to foster long-term talent commitment to the region (even when they live elsewhere).

May 2026, Ireland, Cork	4. Brain Drain and Innovative Solutions	This workshop explores innovative strategies and solutions to mitigate the effects of brain drain while leveraging the opportunities it may create. By embracing innovation, regions can transform the challenge of brain drain into an opportunity to modernize their economies, improve local infrastructure, and create environments that are attractive to both existing residents and returning talent.	To explore and inspire the use of innovative technological, economic, and educational solutions that can both mitigate the negative impacts of brain drain and capitalize on its potential advantages for regional development.
---	---	---	--

6. References

- Artz, G. (2003). Rural area brain drain: Is it a reality? *Choices*, 18(4), 11–15.
- Arundel, R., & Lennartz, C. (2020). Housing market dualization: Linking insider–outsider divides in employment and housing outcomes. *Housing Studies*, 35(8), 1390–1414. <https://doi.org/10.1080/02673037.2019.1667960>
- Beine, M., Docquier, F., & Rapoport, H. (2008). Brain Drain and Human Capital Formation in Developing Countries: Winners and Losers. *The Economic Journal*, 118(528), 631–652. <https://doi.org/10.1111/j.1468-0297.2008.02135.x>
- Berry, R. A., & Soligo, R. (1969). Some Welfare Aspects of International Migration. *Journal of Political Economy*, 77(5), 778–794. <https://doi.org/10.1086/259562>
- Bhagwati, J., & Hamada, K. (1974). The brain drain, international integration of markets for professionals and unemployment: A theoretical analysis. *Journal of Development Economics*, 1(1), 19–42.
- Biedka, W., Herbst, M., Rok, J., & Wojcik, P. (2022). The local-level impact of human capital investment within the EU cohesion policy in Poland. *Papers in Regional Science*, 101(2), 303–326.
- Di Cataldo, M., Monastiriotis, V., & Rodríguez-Pose, A. (2021). How ‘smart’ are smart specialization strategies? *Journal of Common Market Studies*, 59(2), 180–198. <https://doi.org/10.1111/jcms.13156>
- Dobrzanski, P., Bobowski, S., & Clare, K. (2024). Left-behind places in central and eastern Europe—Labour productivity aspect. *Cambridge Journal of Regions, Economy and Society*, 17(1), 137–162.
- Docquier, F., & Rapoport, H. (2012). Globalization, Brain Drain, and Development. *Journal of Economic Literature*, 50(3), 681–730. <https://doi.org/10.1257/jel.50.3.681>
- Faggian, A., Rajbhandari, I., & Dotzel, K. R. (2018). The interregional migration of human capital and its regional consequences: A review. *Transitions in Regional Economic Development*, 227–256.
- Fiorentino, S., Glasmeier, A. K., Lobao, L., Martin, R., & Tyler, P. (2024). ‘Left behind places’: What can be done about them? *Cambridge Journal of Regions, Economy and Society*, 17(2), 259–274. <https://doi.org/10.1093/cjres/rsae012>
- Grubel, H. B., & Scott, A. D. (1966). The international flow of human capital. *The American Economic Review*, 56(1/2), 268–274.
- Haapanen, M., & Tervo, H. (2012). MIGRATION OF THE HIGHLY EDUCATED: EVIDENCE FROM RESIDENCE SPELLS OF UNIVERSITY GRADUATES*. *Journal of Regional Science*, 52(4), 587–605. <https://doi.org/10.1111/j.1467-9787.2011.00745.x>
- MacKinnon, D., Kempton, L., O’Brien, P., Ormerod, E., Pike, A., & Tomaney, J. (2022). Reframing urban and regional ‘development’ for ‘left behind’ places. *Cambridge Journal of Regions, Economy and Society*, 15(1), 39–56.

- Mærsk, E., Sørensen, J. F. L., Thuesen, A. A., & Haartsen, T. (2021). Staying for the benefits: Location-specific insider advantages for geographically *im* mobile students in higher education. *Population, Space and Place*, 27(4), e2442. <https://doi.org/10.1002/psp.2442>
- Mayr, K., & Peri, G. (2009). Brain Drain and Brain Return: Theory and Application to Eastern-Western Europe. *The B.E. Journal of Economic Analysis & Policy*, 9(1). <https://doi.org/10.2202/1935-1682.2271>
- Teney, C. (2021). Immigration of highly skilled European professionals to Germany: Intra-EU brain gain or brain circulation? *Innovation: The European Journal of Social Science Research*, 34(1), 69–92. <https://doi.org/10.1080/13511610.2019.1578197>
- Vazzana, C. M., & Rudi-Polloska, J. (2019). Appalachia Has Got Talent, But Why Does It Flow Away? A Study on the Determinants of Brain Drain from Rural USA. *Economic Development Quarterly*, 33(3), 220–233. <https://doi.org/10.1177/0891242419844320>
- Venhorst, V., Van Dijk, J., & Van Wissen, L. (2011). An Analysis of Trends in Spatial Mobility of Dutch Graduates. *Spatial Economic Analysis*, 6(1), 57–82. <https://doi.org/10.1080/17421772.2010.540033>
- Zhou, Y., Guo, Y., & Liu, Y. (2018). High-level talent flow and its influence on regional unbalanced development in China. *Applied Geography*, 91, 89–98.